

Disorders of the Female Genital Tract in Children and Adolescents: Spectrum and Surgical Management in a Nigerian Paediatric Surgical Practice

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ABSTRACT

Female genital tract disorders in children range from congenital anomalies to acquired and neoplastic conditions. These disorders may affect future reproductive, urinary, and psychosocial function and often require timely surgical intervention. In resource-limited settings, management is challenged by delayed presentation, limited diagnostic facilities, and poor access to multidisciplinary care. This report aimed to describe the spectrum and surgical management of paediatric female genital tract disorders from a paediatric surgical perspective in a resource-constrained setting. A retrospective descriptive study was conducted of female patients aged 0–16 years managed for genital tract disorders between January 2016 and December 2024 at the University of Uyo Teaching Hospital. Seventy-eight patients were included. Congenital anomalies accounted for 36 cases (46.2%), with anorectal malformations and disorders of sexual differentiation being the most frequent. Acquired non-neoplastic conditions (n = 30, 38.4%) included labial adhesions, genital trauma, and clitoral inclusion cysts following female genital mutilation. Eleven neoplastic lesions (14.1%) were recorded, mainly benign ovarian tumours, with some malignant germ cell tumours and vaginal rhabdomyosarcoma. Surgical procedures included adhesiolysis, hymenotomy, vaginoplasty, posterior sagittal anorectoplasty (PSARP), clitoroplasty, and tumour excision. Multidisciplinary involvement was inconsistent, and diagnostic delays were noted in several cases. In conclusion, paediatric surgeons in low-resource settings play a key role in managing a wide range of genital tract disorders in girls, including complex congenital and acquired conditions. Improved early recognition, referral pathways, and access to multidisciplinary care are needed. Further research is warranted to assess long-term functional and psychosocial outcomes.

Keywords: Genital Diseases, Female, Congenital Abnormalities, Urogenital Abnormalities, Gynecologic Surgical Procedures.

INTRODUCTION

Child health is an important component of healthcare delivery worldwide, particularly in low- and middle-income countries (LMICs), where children and adolescents account for a disproportionately large segment of the population. In many sub-Saharan African countries, nearly 50 percent of the population is under 20 years of age, compared to a global average of 35 percent.¹ Paediatric surgical conditions significantly impact child health in these settings yet often remain under-recognised and under-resourced.

Paediatric general surgery involves the diagnosis and management of a broad spectrum of congenital and acquired conditions in neonates, infants, children, and adolescents. These include disorders of the gastrointestinal, thoracic, and genitourinary systems. Among these, conditions affecting the female urogenital tract form a distinct but important subgroup. The evaluation and surgical management of these conditions demand a nuanced understanding of paediatric anatomy, embryology, psychosocial development, and future reproductive potential as well as a commitment to technical precision and gentle tissue handling.²

The reproductive and psychological health of girls aged 0–18 carries long-term implications and deserves focused clinical attention.³ These disorders often cause distress for both the child and their family and may

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influence future fertility and sexual function. Consequently, a multidisciplinary approach is essential, especially one that integrates the expertise of paediatric surgeons, gynaecologists, urologists, endocrinologists, paediatricians, radiologists, and, where necessary, mental health and child protection services.⁴⁻⁷ However, in resource-limited settings, such integrated care is not always accessible, and fragmented referral pathways may delay appropriate management.

There is limited literature from low-resource contexts describing the scope and surgical management of urogynaecological conditions in paediatric surgical practice. Most existing studies are reported from gynaecological rather than paediatric surgical perspectives.⁸⁻¹⁰ This study aims to address this gap by presenting the spectrum and surgical management of paediatric female genital tract disorders encountered in a paediatric surgical unit in a resource-limited setting. The study focuses on the range and frequency of conditions seen, rather than treatment outcomes or long-term follow-up.

MATERIALS AND METHODS

Study Setting and Design

This retrospective, descriptive study was conducted at the University of Uyo Teaching Hospital (UUTH), a 500-bed tertiary referral centre located in Uyo, South-South Nigeria. UUTH provides specialist surgical and medical services and serves as a referral centre for paediatric surgical cases from Akwa Ibom state and neighbouring states. The hospital supports a catchment population of over six million people. Surgical patients who are less than 16 years are usually referred to the paediatric surgery unit. The study covered a 9-year period from January 2016 to December 2024.

Study Population and Eligibility Criteria

The study included all female children aged 16 years and younger who were managed at the Paediatric surgical unit with disorders

related to their genital tracts. Patients with incomplete records and those who did not complete treatment at UUTH were excluded. Given the retrospective nature of the study, a consecutive sampling method was used, where all eligible cases within the study period were included; hence, no formal sample size calculation was required.

Data Collection and Study Variables

Data were extracted from patient case notes and surgical records using a structured data abstraction form developed by the researchers. Extracted information included age, clinical presentation, and surgical procedures. Age was categorised into 0 – 1 year, 2 – 5 years, 6 – 10 years and 11 – 16 years. The major disorders were divided into congenital, acquired and neoplastic.

Data Analysis

Statistical analysis was conducted using Excel 365 for Windows. Descriptive statistics were presented as medians with Interquartile ranges as well as counts and percentages.

Ethical Considerations

This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. Institutional approval was obtained from the Health Research Ethics Committee of the University of Uyo Teaching Hospital prior to data collection (Protocol number: UUTH/AD/S/96/VOL. XXII/009). Given the retrospective nature of the study, the use of anonymised patient data did not require individual informed consent. However, informed consent was obtained from parents and guardians for the publication of anonymised clinical photographs included in this manuscript. All data were handled with strict confidentiality, and no identifying patient information was disclosed.

RESULTS

Age Distribution and Clinical Presentation

Over the 9-year period, a total of 78 paediatric patients were managed for female

urogenital tract disorders. The median age was 3 years (IQR: 1–7 years). The highest proportion were infants aged 0–1 year (39.7%), followed by children aged 2–5 years (28.2%). Full age distribution is shown in Table 1.

Table 1 Age Distribution of Patients Managed for Female Genital Tract Disorders in the Paediatric Surgical Unit (n = 78); Median age: 3 years (IQR: 1–7 years)

Age Group (years)	Number of Cases	Percentage
0 – 1	31	39.7
2 – 5	22	28.2
6 – 10	11	14.1
11 – 16	14	18.0
Total	78	100

Congenital Abnormalities

Congenital anomalies accounted for 36 of the 78 cases, with abnormalities associated with Anorectal Malformation and Disorders of Sexual Differentiation accounting for the most frequent with 16 cases each. Most of these anomalies required surgical intervention, ranging from vaginoplasty to clitoroplasty and posterior sagittal anorectoplasty (PSARP). Details are summarised in Table 2.

Acquired Non-Neoplastic Abnormalities

A total of 30 patients presented with acquired disorders of the female genital tract. Labial adhesions (n=12, 15.4%) were the most common and were managed with adhesiolysis. Non-neoplastic swellings were also excised in 11 patients (14.1%). Table 3 summarises the disorders and interventions.

Neoplastic Lesions

Eleven neoplastic lesions were recorded, the majority of which were ovarian tumours. These included four benign and three

malignant ovarian tumours. All patients underwent surgical excision, and those with malignant disease received adjuvant chemotherapy. The distribution is presented in Table 4. Representative clinical images of selected cases are shown in Figure 1.

DISCUSSION

Disorders of the female urogenital tract in children are important clinical conditions that often require multidisciplinary collaboration. Special attention must be given to preserving the patient's future sexual, reproductive, and psychological health. Due to the unique anatomical and physiological considerations in this age group, many of the younger children are referred by the paediatricians to paediatric general surgeons for initial evaluation and management.

This study aimed to highlight and describe the spectrum of female genital tract disorders encountered in children and adolescents managed within a paediatric surgical unit at a tertiary referral hospital. Our findings indicate that congenital anomalies were the most frequently managed group, with anorectal malformations involving the urogenital tract being the most common congenital abnormality. These were followed by disorders of sexual differentiation and a range of acquired conditions, including inflammatory, infectious, traumatic, and neoplastic lesions. The diversity of presentations highlights the role of paediatric surgeons in managing both congenital and acquired gynaecological disorders, particularly in settings where specialised paediatric gynaecology services may be limited.

Congenital anomalies constituted the largest group of cases in this study, accounting for nearly half of all female genital tract disorders requiring paediatric surgical consultation. This finding is consistent with reports from other paediatric surgical settings, where congenital abnormalities are the predominant indication for surgical

Table 2 Summary of Congenital Female Genital Tract Abnormalities by Category, Disorder, Frequency, and Surgical Intervention (n = 36)

Category	Disorder	Number of Cases	%	Surgical Procedure Performed
▪ <i>Mullerian Abnormalities</i>	a. Bicornuate Uterus	1	1.3	No surgical intervention
	b. Unicornuate Uterus	1	1.3	No surgical intervention
	c. Vaginal Atresia	2	2.6	Vaginoplasty
▪ <i>Abnormalities of the Urogenital Sinus, Cloaca and External Genitalia</i>	a. Clitoromegaly/Ambiguous Genitalia (DSD)	7	9.0	Clitoroplasty ± Feminising Genitoplasty
	b. Persistent Urogenital Sinus	2	2.6	Urogenital Sinus Mobilisation
	c. Imperforate Hymen	5	6.4	Hymenotomy
	d. Bladder Exstrophy	2	2.6	CPRE
▪ <i>Abnormalities Associated with Anorectal Malformations</i>	a. Persistent Cloaca	4	5.1	PSARP ± Urogenital Reconstruction
	b. ARM with Recto-vaginal/Recto-vestibular Fistula	12	15.4	PSARP

Note: DSD = Disorders of Sexual Differentiation; ARM = Anorectal Malformation, CPRE = Complete Primary Repair of Exstrophy; PSARP = Posterior Sagittal Anorectoplasty.

intervention, with reported rates ranging up to 88 percent.¹¹⁻¹⁵ In contrast, other studies from gynaecological centres within Nigeria have identified septic abortion and menstrual disorders as the most common presentations in children and adolescents.⁸⁻¹⁰ These differences likely reflect the variation in case mix between paediatric surgical and gynaecological practice settings.

Anorectal malformations were the most common congenital abnormalities in our study, consistent with reports from other centres.¹⁶ Their embryological link to the urogenital tract accounts for the frequent coexistence of reproductive anomalies. Rectovestibular fistula is the most common defect in girls and generally carries a favourable prognosis after surgical correction.^{17,18} In contrast, persistent cloaca represents a more complex spectrum,

involving a single perineal opening for the rectum, vagina, and urethra, due to incomplete division of the cloaca by the urorectal septum. The diagnosis is clinical and should be suspected in any female neonate with an imperforate anus and apparently small external genitalia.¹⁹ In our cohort, these patients were managed using standard staged protocols, including an initial colostomy followed by definitive Posterior sagittal anorectoplasty (PSARP) or Posterior sagittal anorecto-vagino-urethroplasty (PSARVUP) from six months of age. Although long-term outcomes were not assessed, other studies have reported satisfactory continence and reproductive outcomes when appropriate follow-up and supportive care are available.²⁰⁻²²

Although Disorders of Sexual Differentiation (DSD) are less frequently discussed in general paediatric surgical

Table 3 Acquired Non-neoplastic Disorders of the Female Genital Tract Managed in the Paediatric Surgical Unit, Categorised by Aetiology, Frequency, and Surgical Intervention (n = 30)

Category	Specific Disorder	Number of Cases	%	Surgical Procedure Performed
<i>Inflammatory/Infectious conditions</i>	Labial adhesion	12	15.4	Adhesiolysis
	Septic Abortion	2	2.6	Exploratory laparotomy and Evacuation of retained products of conception
<i>Post-traumatic conditions</i>	Anovaginal fistula	1	1.3	Vaginoplasty/Fistula repair
	Rectovaginal fistula	1	1.3	Vaginoplasty/Fistula repair
	Vaginal laceration	1	1.3	Vaginal Repair
	Vaginal stenosis	2	2.6	Vaginal dilatation
<i>Non-neoplastic growths</i>	Clitoral cyst	3	3.8	Excision ± deinfibulation
	Angiokeratoma with fibrolipoma of the labia	1	1.3	Excision
	Urethral prolapse	6	7.7	Excision
	Urethral polyps	1	1.3	Excision

Table 4 Neoplastic Lesions of the Female Genital Tract and Pelvis Managed in the Paediatric Surgical Unit, Categorised by Tumour Type, Anatomical Origin, and Frequency (n = 11)

Tumour type	Origin	Specific tumour	Number of Cases (n)	%
<i>Benign</i>	Ovarian	Mature Teratoma	2	2.6
		Serous cystadenoma	2	2.6
<i>Malignant</i>	Ovarian	Cyst adenocarcinoma	1	1.3
		Immature teratoma	1	1.3
		Mixed germ cell tumour	1	1.3
	Vaginal	Vaginal botryoid Rhabdomyosarcoma	1	1.3
	Sacroccocygeal region (Pelvic extension)	Malignant Sacroccocygeal teratoma	2	2.6

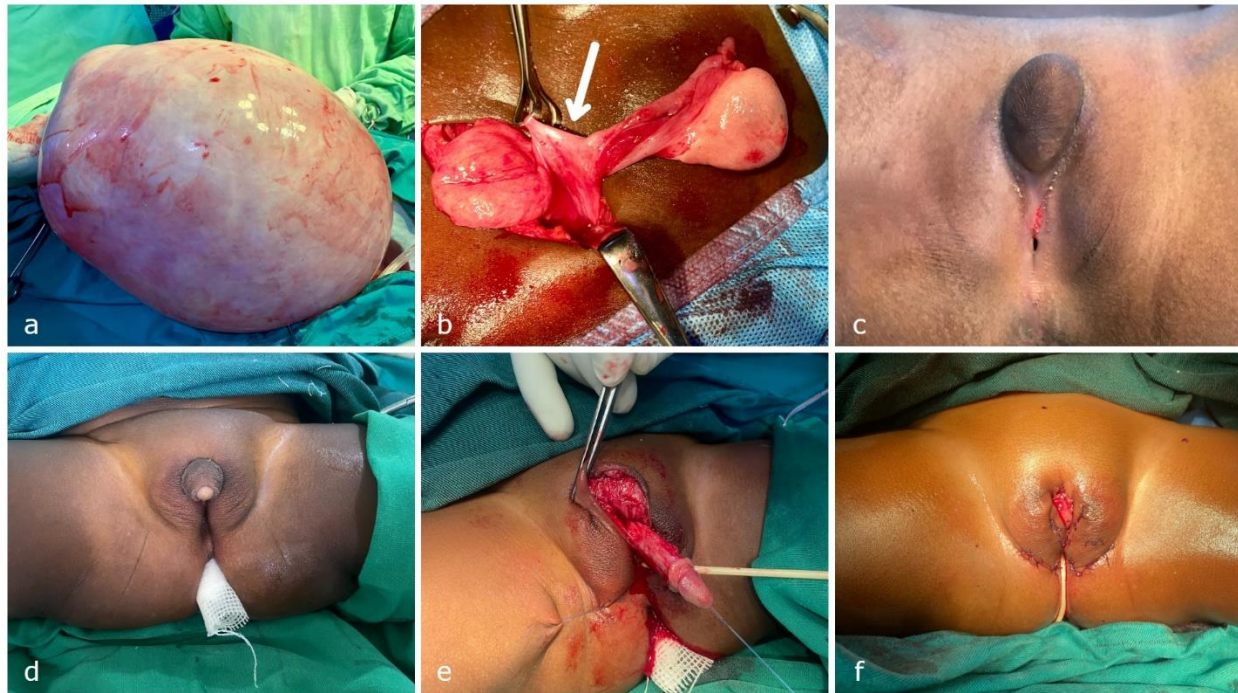


Figure 1 a) Giant malignant germ cell tumour involving the entire abdominal cavity of a 12-year-old girl. b) Unicornuate uterus (white arrow) and small mature teratoma (dermoid cyst) in a 4-year-old. c) Clitoral inclusion cyst in an 8-year-old with Type III Female Genital Mutilation. d) A 1-year-old with Congenital Adrenal Hyperplasia and clitoromegaly and subsequent clitoroplasty and feminising genitoplasty (e and f)

literature, they have been identified in dedicated urogenital anomaly studies as important contributors to surgical workload.^{23,24} The approach to managing these conditions, particularly the timing and extent of surgical intervention, remains a subject of ongoing clinical and ethical debate.²⁵ Each case warrants individualised assessment and management. Key considerations include the risk of gonadal tumours, potential for future fertility, susceptibility to urinary tract infections, avoidance of stigma related to atypical genital anatomy, and preservation of functional genital anatomy to allow for future penetrative intercourse.^{25,26} In high-income countries, multidisciplinary DSD clinics typically coordinate diagnosis, endocrinologic assessment, psychosocial support, and surgical intervention.^{4,26,27} In contrast, such coordinated care is often lacking in low-resource settings, resulting in delayed presentation and fragmented management.²⁴

In our cohort, surgical intervention was performed early in girls with congenital adrenal hyperplasia, in line with reports suggesting favourable outcomes following early genital reconstruction.^{26,28} The proportion of Müllerian anomalies in our series was relatively low, possibly reflecting their tendency to present later in adolescence or early adulthood, and their more frequent referral to the obstetrics and gynaecology units rather than paediatric surgical services.²⁹

Among the acquired disorders, labial adhesions were the most frequently encountered in our series. Randawa *et al.*,³⁰ in a study from Northern Nigeria, similarly documented labial adhesions as the most common condition among paediatric gynaecology patients. These adhesions are often secondary to chronic vulvovaginitis, particularly in prepubertal girls, where low oestrogen levels, poor hygiene, and recurrent

local irritation predisposes to epithelial denudation and adhesion formation.⁷ Most cases respond well to simple adhesiolysis, although recurrence may occur if underlying inflammation is not adequately addressed.^{7,30} None of our patients were given topical oestrogen therapy because it was not readily available.

Three patients with clitoral inclusion cysts resulting from female genital mutilation (FGM) were also observed in our cohort. This practice, regrettably, persists in certain communities despite the Violence Against Persons (Prohibition) Act of 2015, which prohibits FGM in Nigeria.³¹ Onah *et al.*³² found that, despite the legal ban, women who had been circumcised were more likely to have circumcised their daughters and intended to do so again in the future. The exact incidence of clitoral inclusion cysts has not been determined. However, several authors have reported their occurrence in children.³³⁻³⁶ Surgical excision remains the mainstay of treatment, along with deinfibulation or scar release, particularly in cases involving fused vaginal openings.³⁷

Traumatic lesions, though less frequently encountered, carry substantial psychosocial implications. Studies have shown that perineal injuries represent approximately 0.2 percent of all injuries in children aged 15 years and younger.²⁹ In prepubertal girls, the hypoestrogenised genital tissues are less distensible and more fragile, making them particularly susceptible to both accidental and non-accidental trauma.²⁹ In our series, a six-year-old presented with passage of stool per vaginam four days after being sexually assaulted by an adult male. Similarly, Randawa *et al.*³⁰ reported assault-related genital injuries in nine girls under 12 years of age. We also managed a one-year-old who developed an anovaginal fistula following a traumatic enema. Both patients underwent staged surgical repair, which included diverting colostomies and subsequent vaginoplasties.

Ovarian tumours comprised the majority of neoplastic lesions in this study, most of which were benign. Mature cystic teratomas were the most common histological subtype, consistent with findings from Ajani *et al.*,³⁸ and typically occurred in adolescents. Conservative surgical approaches such as ovarian cystectomy were used where feasible to preserve ovarian function. However, oophorectomy was required in cases of torsion with ischaemia, suspected malignancy, or giant tumours where anatomical dissection was difficult. Giant ovarian tumours, defined as measuring 15 centimetres or more in diameter, pose difficulties for ovary-sparing procedures due to distortion of normal anatomy, difficulty distinguishing cyst walls from compressed ovarian tissue, and the need for larger, more invasive incisions.^{39,40,41,42} Diagnostic delays were common, with some cases initially misdiagnosed due to overlapping abdominal symptoms. Torsion of ovarian cysts frequently presented as an acute abdomen, while some pelvic masses were misdiagnosed as originating outside the pelvis. These diagnostic challenges emphasise the importance of maintaining a high index of suspicion when evaluating female children presenting with abdominal pain or masses.

In addition to ovarian pathology, our cohort included rare cases of vaginal botryoid rhabdomyosarcoma and malignant sacrococcygeal teratoma with pelvic extension. These tumours are infrequently encountered in paediatric surgical practice and are rarely represented in most surgical series.⁴³ Randawa *et al.*³⁰ similarly reported a single case of botryoid rhabdomyosarcoma, further illustrating their rarity. All malignant tumours in our study had surgical excision and were co-managed with the paediatric oncology team for adjuvant therapy.

The management of paediatric female genital tract disorders in our setting often required collaboration across specialties. Many of our patients benefitted from multidisciplinary input involving paediatric

surgery, urology, paediatrics, endocrinology, gynaecology and, where necessary, oncology and social welfare services. For example, children with malignant tumours were co-managed with the paediatric oncology team, while those involving sexual assault and post-FGM complications required both surgical intervention and psychosocial support. However, multidisciplinary care was not consistently available for all patients, largely due to systemic limitations. For example, Bladder exstrophy-epispadias complex, though rare, represents an unmet need. Its multisystem complexity, the requirement for specialised surgical teams, and the lifelong social and economic challenges faced by affected families illustrate the broader gaps in coordinated care.⁴⁴

This study has some limitations. As a retrospective hospital-based review, it is subject to documentation bias and incomplete clinical details, particularly in older records. The analysis was restricted to cases managed within the paediatric surgical unit, which may not fully reflect the broader spectrum of paediatric gynaecological conditions seen in other departments such as gynaecology or paediatrics. In addition, detailed long-term follow-up data on functional, reproductive, and psychosocial outcomes were not available for most patients, limiting our ability to comment on the long-term effectiveness of surgical outcomes, particularly for complex conditions. However, it is important to emphasise that the primary objective of this study was not to evaluate surgical outcomes, but rather to describe the types of female genital tract disorders encountered in our paediatric surgical practice and the operative interventions undertaken. A prospective follow-up study is planned to evaluate these outcomes more comprehensively and to guide improvements in long-term multidisciplinary care.

Conclusion

This study demonstrates the important role paediatric surgeons play in the evaluation and surgical management of a diverse range of female genital tract disorders, including congenital anomalies, acquired conditions, and neoplastic lesions. In resource-limited settings, the complexity of care is heightened by delayed presentation, restricted access to specialised investigations, and inconsistent multidisciplinary support. Early recognition and timely collaborative care are essential to improving outcomes. Strengthening referral systems and improving awareness among primary care providers may help to reduce diagnostic delays. While this study provides an overview of the spectrum of disorders encountered in paediatric surgical practice, further prospective research with long-term follow-up is needed to evaluate functional, reproductive, and psychosocial outcomes more comprehensively.

Conflicts of Interest: None declared

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